GOVERNMENT OF PAKISTAN | CONSTITUENCY ADMINISTRATION

CHRAS

Constituency Human Resource & Allocation System

INTERNAL HR SYSTEM · ROLE, CAPABILITY & WORKFLOW DOCUMENTATION

PREPARED FOR	DOCUMENT	VERSION
Client Review	Product & Operations Guide	1.0 | October 2026

STATUS

Working Demo / Internal Use

Confidential — for client review and internal planning

01 / INTRODUCTION

Contents & Executive Summary

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CHRAS is a web-based internal HR platform for constituency-level appointments. It brings candidate registration, document verification, seat allocation, official appointment letters, reporting and internal coordination into one controlled system.

10	9	7	12
LOGIN TYPES	OFFICER ROLES	WORKFLOW STEPS	DISTRICTS SEEDED

Motive in one sentence: Replace scattered registers and manual files with one clear system where every candidate is registered, checked, appointed and tracked — and no one can hold two active seats by mistake.

What the system delivers

01

One candidate record

Personal details, CNIC, academics, photograph and allocation history stay in one place.

02

Controlled authority

Each role sees only the modules and buttons allowed by its permissions.

03

Transparent allocation

District and constituency quotas, duplicate-allocation blocking and documented revocation.

04

Candidate self-service

A read-only portal shows the candidate their own status, seat and appointment letter.

Document scope: This guide describes the current working build. It is written for a client walkthrough, role confirmation and acceptance discussion. Names shown are demo accounts and can be replaced with real officers during deployment.

02 / PURPOSE

The Problem and the Solution

Without a system

- Candidate files live in registers, Excel sheets and paper folders.
- Officers cannot quickly see who is verified, allocated or pending.
- The same person can be appointed to two active roles by mistake.
- Candidates have no clear way to check their own status.
- Reports require manual counting and take hours to prepare.
- There is no consistent record of who cancelled an appointment and why.

With CHRAS

- Every candidate has one digital record from enrolment onward.
- Dashboard cards and filters show verified, pending and allocated counts.
- The system blocks a second active allocation for the same candidate.
- Candidates log in and view their own seat and appointment letter.
- Reports are generated from live data in a few clicks.
- Revocation requires a written reason and automatically frees the seat.

Simple example: A candidate registers, an authorised officer verifies the record, an allocator assigns a seat, and the system creates the appointment reference. If the appointment is later cancelled, the reason is stored and the candidate returns to the verified pool.

Who uses the platform

A

Administration

Admin, MNA, MPA, District Chairperson, Civil Officer and Committee Head.

B

Field and office staff

UC Chairman, Councilor and Public Sector Staff handle enrolment and record entry.

C

Candidates

Citizens with a read-only portal for their own record, allocation and letter.

Design principles

Role-based access

A staff worker can register a candidate but cannot allocate a seat unless the role allows it.

One source of truth

Dashboard, reports, allocation and candidate portal all read the same stored records.

Documented decisions

Allocations carry a reference number; cancellations carry a reason and date.

Read-only candidate view

Candidates can see their own information but cannot change official records.

03 / PEOPLE

Roles, Names and Capabilities

The names below are demo accounts inside the current build. In deployment they represent the client's real officers.

#	ROLE / DEMO PERSON	MAIN WORK	WHAT THEY CAN DO	WHAT THEY CANNOT DO
1	**Admin** Dr. Tariq Jamil Chishti FULL CONTROL	Overall system control and oversight.	Register and verify candidates, allocate seats, cancel appointments, generate reports, send messages, view all records.	No restrictions in the current build.
2	**MNA** Malik Naveed Awan ELECTED	Constituency oversight and appointments.	Register and verify candidates, allocate seats, generate reports, send messages.	Cannot cancel an appointment; cannot change system-wide settings.
3	**MPA** Syeda Fatimah Bokhari ELECTED	Provincial oversight and appointments.	Register and verify candidates, allocate seats, generate reports, send messages.	Cannot cancel an appointment; cannot change system-wide settings.
4	**District Chairperson** Chaudhry Riaz Gujjar DISTRICT	District-level hiring and administration.	Register and verify candidates, allocate seats, cancel appointments, generate reports, send messages.	No district-management screen; no system-wide settings.
5	**Civil Administration Officer** Kamran Shahbaz CIVIL SERVICE	Civil appointments and official letters.	Register and verify candidates, allocate seats, cancel appointments, generate reports, send messages.	No restriction on the normal HR workflow.
6	**Committee Head** Prof. Dr. Nargis Farooq AUDIT	Independent audit and review.	Review candidate records, generate audit reports, send messages.	Cannot register candidates; cannot allocate seats.
7	**UC Chairman** Haji Abdul Rehman Durrani LOCAL	Grassroots enrolment.	Register and verify local candidates, send messages.	Cannot allocate seats, cancel appointments or generate reports.
8	**Councilor** Bilal Hassan Abbasi LOCAL	Ward-level enrolment.	Register and verify local candidates, send messages.	Cannot allocate seats, cancel appointments or generate reports.
9	**Public Sector Staff** Zubair Ahmad Qazi STAFF	Data entry and record checking.	Register and verify candidate records, send messages.	Cannot allocate seats, cancel appointments or generate reports.
10	**Candidate** Registered citizen, e.g. Muhammad Hamza Tariq READ-ONLY	View own record and appointment.	See own profile, status, allocated seat and appointment letter.	Cannot edit records, view other candidates, allocate or cancel anything.

Important distinction: being an office worker does not automatically grant allocation rights. Zubair Ahmad Qazi (Staff) can register a candidate but cannot give a seat. Chaudhry Riaz Gujjar (District Chairperson) can register, allocate and cancel. Authority follows the role, not the desk.

04 / WORKFLOW

Complete End-to-End Flow

1 Apply or register

The candidate signs up online, or an authorised officer registers them. CNIC, education, photograph and portal password are saved. **Candidate, Staff, UC Chairman, Councilor, MNA, MPA, District Chairperson, Civil Officer, Admin.**

2 Check and verify the record

The enrolling officer checks identity and documents. Officer enrolment saves the record as **Verified**; self-signup starts as **Registered**. **Registering roles above; Committee Head can review but not edit.**

3 Allocate a seat or role

The allocator selects the candidate, district, constituency, role and tenure. The system blocks a second active seat for the same candidate. **Admin, MNA, MPA, District Chairperson, Civil Officer.**

4 Issue the appointment letter

The system creates an official appointment reference and Gazette-style letter, and marks the candidate **Allocated**. **System; allocator can view the letter.**

5 Candidate views the result

The candidate logs in to a read-only portal and sees their own record, allocated seat and appointment letter. **Candidate only.**

6 Cancel with a reason, if needed

The authorised officer writes a reason. The appointment becomes **Revoked**, the candidate returns to **Verified**, and the seat is freed. **Admin, District Chairperson, Civil Officer.**

7 Report and coordinate

Dashboards show quota use, verification progress and role/party breakdowns. Reports can be generated and staff can message each other. **Authorised report roles and all officer roles.**

05 / PERMISSIONS

Permission Matrix

ROLE	REGISTER / VERIFY	ALLOCATE SEAT	CANCEL	REPORTS	MESSAGES	OWN PORTAL
Admin	✓	✓	✓	✓	✓	—
MNA	✓	✓	—	✓	✓	—
MPA	✓	✓	—	✓	✓	—
District Chairperson	✓	✓	✓	✓	✓	—
Civil Officer	✓	✓	✓	✓	✓	—
Committee Head	—	—	—	✓	✓	—
UC Chairman	✓	—	—	—	✓	—
Councilor	✓	—	—	—	✓	—
Public Sector Staff	✓	—	—	—	✓	—
Candidate	—	—	—	—	—	✓

This matrix reflects the current user-facing build. Buttons and tabs are hidden or blocked when a role lacks permission.

06 / PRODUCT

Modules and Technology

01

Public landing portal

Government-style introduction, district quota cards, role information, CNIC verification lookup and registration calls to action.

02

Authentication

Separate officer and candidate sign-in, demo role selection, secure candidate password hashing and session handling.

03

Analytics dashboard

Total registered candidates, active allocations, quota utilisation, verification progress, role and party breakdowns.

04

Candidate registration

Personal details, CNIC, domicile, party affiliation, matric/FSc/higher education and photograph upload preview.

05

Seat allocation

Allocation form, duplicate-seat prevention, searchable appointment register, Gazette letter view and reasoned revocation.

06

Appointment letters

Official reference number, appointee details, role, district, tenure, issuing authority and print-ready layout.

07

Reports

District summaries, party and role breakdowns, verified allocation table and press-release copy action.

08

Internal messaging

Role-prefilled official notices, priority levels, district filters and read/unread tracking.

09

Candidate portal

Read-only personal record, academic history, allocation details and appointment letter access.

Application routes

ROUTE	PURPOSE	ACCESS
/	Landing page and authenticated dashboard shell	Public; dashboard after login
/login	Officer and candidate sign-in	Public
/signup	Candidate self-registration	Public
/api/auth/candidate	Candidate credential verification	API
/api/candidates	Candidate list, create and update records	API
/api/allocations	Appointment list and seat allocation	API
/api/analytics	Dashboard totals and breakdowns	API

Technology stack

Frontend

Next.js 16 App Router, React 19, TypeScript, Tailwind CSS v4.

UI and feedback

lucide-react icons, Sonner notifications, canvas-confetti confirmations.

Backend and data

Next.js route handlers, Node crypto scrypt, JSON file database.

Design language: Pakistan green, white and restrained gold. Green is reserved for official actions and identity; party and role badges use distinct colours so the dashboard remains readable.

07 / API & DATA

API Surface, Data Model and Security

ENDPOINT	METHODS	PURPOSE
/api/candidates	GET, POST	List candidates; create a candidate with hashed password.
/api/candidates/[id]	GET, PATCH, DELETE	Read, update or remove one candidate; password hashes are stripped from responses.
/api/allocations	GET, POST	Filter allocation register; allocate a seat and generate an appointment reference.
/api/allocations/[id]	PATCH	Revoke an allocation with a reason; frees the seat and restores candidate status.
/api/analytics	GET	Dashboard totals with optional district filter.
/api/districts	GET	District list, quotas and constituency data.
/api/messages	GET, POST, PATCH	Read messages, send notices, mark messages as read.
/api/auth/candidate	POST	Verify CNIC and candidate portal password.

Core records

- **Candidate:** identity, contact, domicile, party, academics, documents, status and allocated role.
- **District:** province, constituencies, total quota and allocated count.
- **Allocation:** reference number, candidate, role, district, constituency, tenure, issuing authority and status.
- **Message:** sender, role, district, subject, body, priority and read state.

Status values

- **Candidate:** Registered, Verified, Allocated, Under Review.
- **Allocation:** Active, Revoked, Completed.
- **Priority:** Urgent, Official Notice, Routine, Inquiry.

Security and production readiness

Implemented now: candidate passwords are hashed with scrypt and salts; password hashes never leave the server; sessions are stored locally; duplicate active allocations are blocked; revocation requires a stored reason.

Before real government deployment: replace demo officer passwords with real authentication/SSO; add server-side session and role enforcement on every API; move from JSON file storage to PostgreSQL or SQL Server; connect real document storage and NADRA verification; add audit logs, backups, HTTPS and access monitoring.

Why this matters: the current build is complete as a working demo and internal review system. The items above are production hardening, not missing screens in the normal user flow.

08 / HANDOVER

Run Guide, Acceptance Checklist and Next Steps

Run the project

```
# install dependencies
npm install

# start development server
npm run dev

# quality checks
npx tsc --noEmit
npm run lint
npm run build

# open in browser
http://localhost:3000
```

Demo login

LOGIN TYPE	USERNAME	PASSWORD
Admin	**admin**	admin
MNA	**mna**	mna
MPA	**mpa**	mpa
District Chairperson	**chairperson**	chairperson
UC Chairman	**uc_chairman**	uc_chairman
Councilor	**councilor**	councilor
Civil Officer	**civil_officer**	civil_officer
Committee Head	**committee_head**	committee_head
Staff	**staff**	staff
Candidate	CNIC	CNIC digits (demo) or chosen password

Client acceptance checklist

- ☐ Public landing page and candidate verification lookup
- ☐ Candidate self-registration with portal password
- ☐ Candidate registration with academic record and photograph
- ☐ Appointment letter and Gazette reference
- ☐ Dashboard analytics, quota and party/role breakdowns
- ☐ Internal messaging with read/unread state
- ☐ Officer and candidate sign-in routes
- ☐ Role-based sidebar, tabs and action buttons
- ☐ Seat allocation with duplicate-seat prevention
- ☐ Reasoned appointment cancellation
- ☐ Reports and press-release copy action
- ☐ Read-only candidate portal

Recommended next steps

1. Confirm role policy
Client approves who may register, allocate, cancel and report.

2. Replace demo accounts
Create real officer accounts and remove demo passwords.

3. Move to production data
Database, backups, document storage and server-side permissions.

4. Pilot one district
Run a controlled pilot, collect feedback, then roll out province-wide.

Summary for the client

CHRAS gives every role a clear boundary. Staff can register and check records. Authorised officers can allocate and cancel seats. Auditors can review and report. Candidates can only view their own appointment. Every action stays connected to one candidate record, one allocation register and one audit-friendly history.

Confirmation point: the current build treats officer enrolment as verification. Self-registered candidates start as *Registered*. If the client requires a separate second approval step, confirm that before final acceptance.